

Micro Energy Harvesting 2026

Exercise 5: Capacitors, batteries, efficiencies and leakage currents

Task 1: Radio Messages

You want to supply a radio system from a charged capacitor. For the transmission of a protocol including running the microcontroller you need 200 μJ . The MC is connected via a two-point switch at a capacitor voltage of 2.5 V and disconnected again at 1.9 V. What is the minimum capacity of the capacitor?

Task 2: NiMH Batteries

A NiMH battery has a Coulomb efficiency η_C of 68% and an energy efficiency η_E of 59%. Its end-of-discharge voltage is 1.0 V and its end-of-charge voltage is 1.5 V. The capacity C_0 of the cell is 15 mAh, the self-discharge rate SDR is 0.2 %/day.

- a) How many hours can you run a low power system continuously on this battery, having the following properties, assuming that no self-discharge occurs during this period:
Operating voltage (kept constant in the system): 1,0 V
Operating power (constant): 150 μW
- b) What current would an energy harvesting system have to provide to at least compensate for the continuous self-discharge of the battery? Note the corresponding Efficiency.
- c) What charge quantity Q is required to fully charge the battery once?
- d) What current would an energy harvesting system have to provide to keep the battery charged and to supply the low power system continuously. The low power system is supplied directly from the generator. What would change if it was also fed from the battery?

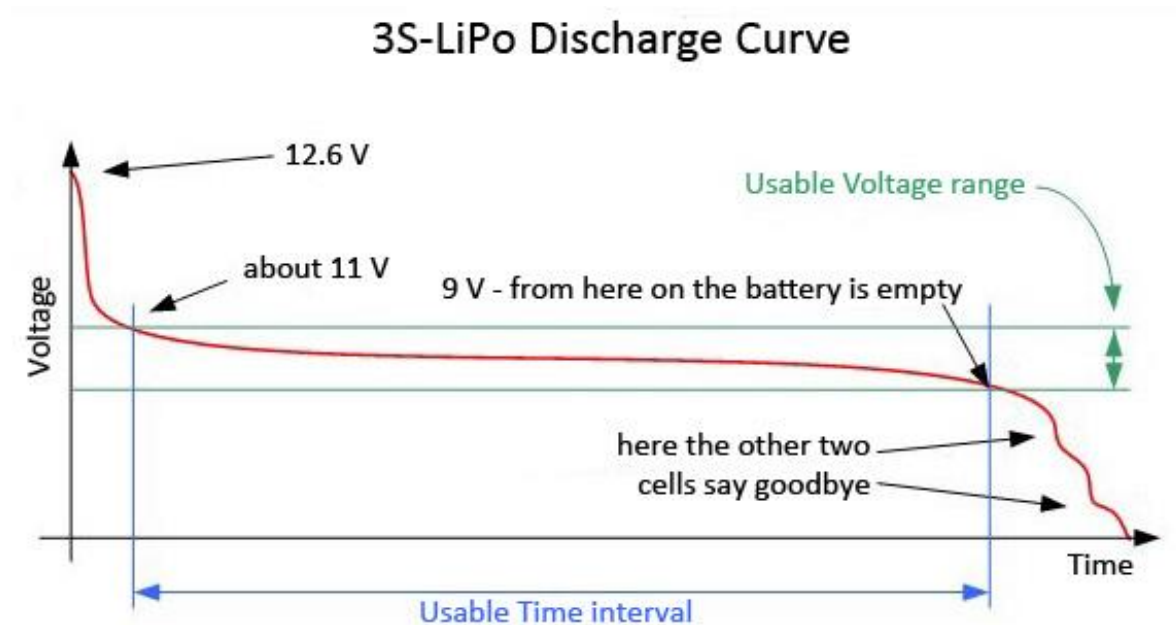
Task 3: Li-Ion Battery

A lithium-ion battery has a Coulomb efficiency η_C of 98 % and an energy efficiency η_E of 90 %. Its end-of-discharge voltage is 3.3 V, its end-of-charge voltage 4.1 V. In this range the battery behaves linear (C-V). The capacity C_0 of the cell is 1.5 Ah, the self-discharge rate SDR is 1.9 %/month (30 d).

- a) Determine the equivalent electrical resistance (EPR) of the battery from the self-discharge rate and the capacity.
- b) What current and power would an energy harvesting generator have to provide to at least compensate for the continuous self-discharge of the battery?
- c) How much energy is required to fully charge the battery once?

Task 4: LiPo-Battery with three cells

The picture below shows the discharge curve of a three cell LiPo battery. A single cell has a usable voltage between 3.8 and 4.0 V. How would you determine the state of charge of the battery?



Why are Li-Ion batteries and Li-Po better than NiMH batteries? How does it compare to non-rechargeable batteries? See below.

